Experimental Calibration of a Superconducting Qubit with QICK

Summary sheet based on the measurement notebook `06_qubit_2026_06_17`

Introduction

This note summarizes, in the same order as in the notebook, the different calibration and characterization steps performed on the qubit. The goal is not only to describe the programs, but also to explain what each measurement is used for, how the results are interpreted, and why each calibrated parameter is useful for the next experiments.

1 Readout-chain verification: *ADC*

The first step is to verify that the readout chain works properly and to set two essential parameters:

- the resonator readout frequency f_{res};
- the ADC delay, namely the time at which the digital integration window starts.

The `LoopbackProgram` sends a readout pulse through the resonator line, without exciting the qubit, and then measures the response on the ADCs. The `SingleToneSpectroscopyProgram` is used to scan the resonator frequencies when the qubit is in the ground state $|g\rangle$.

How to choose the readout frequency

The frequency is scanned around the resonator resonance and the measured signal amplitude is monitored. The best frequency is the one that gives the largest response, or more precisely the best contrast between the responses associated with the two qubit states.

In practice, this frequency will be reused in all the following experiments.

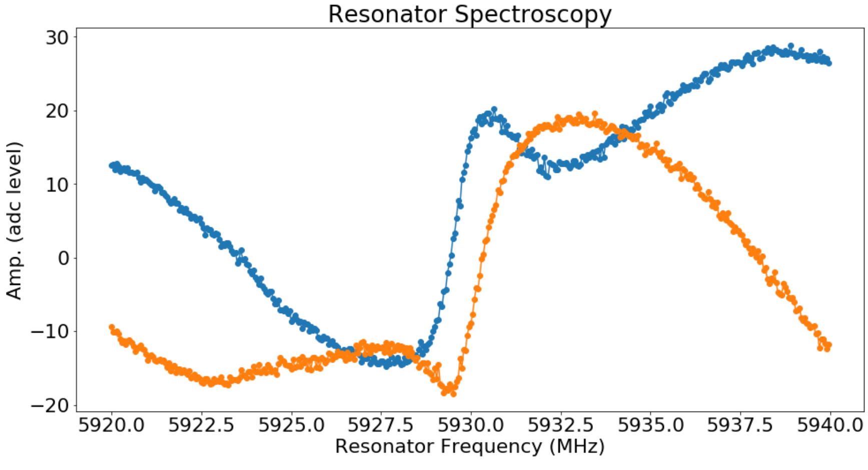

Figure 1: Resonator spectroscopy used to determine the optimal readout frequency.

How to choose the ADC delay

The ADC delay (`adc_trig_offset`) sets the beginning of the digital acquisition window relative to the analog readout pulse. The parameter is chosen by observing where the readout signal becomes stable:

- if the trigger is too early, the rising edge or transient response is integrated;
- if the trigger is too late, part of the useful signal is missed.

The best ADC delay is therefore the one that places the integration window in the middle of the region where the resonator response is the cleanest and most stable.

This parameter is crucial for the rest of the notebook: all population measurements rely on a reliable readout. A poor ADC delay directly degrades the quality of the following experiments.

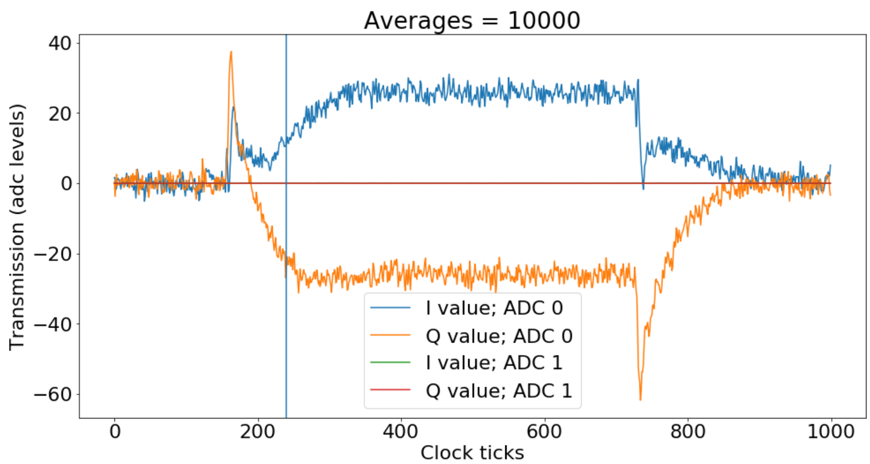

Figure 2: Loopback measurement used to optimize the readout timing.

2 Qubit spectroscopy

To find the qubit transition frequency f_{ge}, the `PulseProbeSpectroscopyProgram` is used.

A fixed-duration qubit pulse is applied while its frequency is scanned. After each pulse, the qubit state is read out through the resonator.

Interpretation

When the drive frequency is close to the transition $|g\rangle \leftrightarrow |e\rangle$, a dip or a peak appears on the spectroscopy curve because of resonance.

From this measurement, the qubit transition frequency is extracted:

$$f_{ge}.$$

This frequency is then used in all later qubit-control pulses.

3 Calibration of π and $\pi/2$ pulses: Rabi oscillations

The next step is to determine the pulse gains corresponding to rotation pulses:

`pi_gain` and `pi2_gain`.

Two Rabi experiments are used in the notebook:

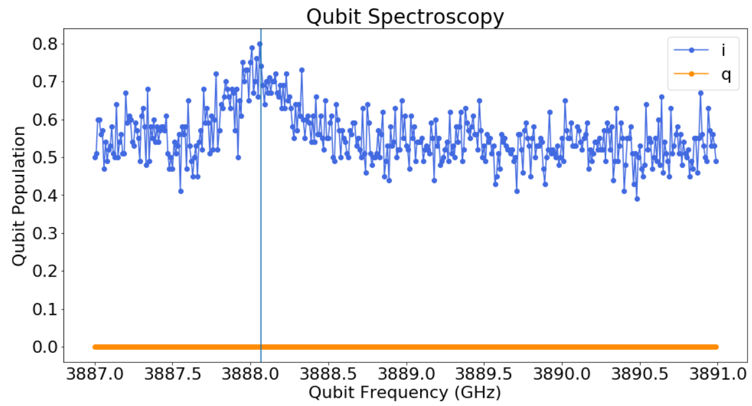

Figure 3: Qubit spectroscopy after threshold calibration.

- **Amplitude Rabi**: the pulse duration is fixed and the gain is swept;
- **Length Rabi**: the gain is fixed and the pulse duration is swept.

Both approaches measure the same physical quantity: the pulse area, which sets the rotation angle on the Bloch sphere.

Physical interpretation

When the gain is swept, the rotation angle increases gradually and the excited-state population oscillates:

$$P_e(G) \simeq \sin^2\left(\frac{\alpha G}{2}\right).$$

This oscillation is the standard Rabi oscillation of a resonantly driven two-level system. From this curve, one identifies:

- the first maximum as the π pulse;
- half of that rotation angle as the $\pi/2$ pulse.

These two values are fundamental for all coherent-control experiments on the qubit.

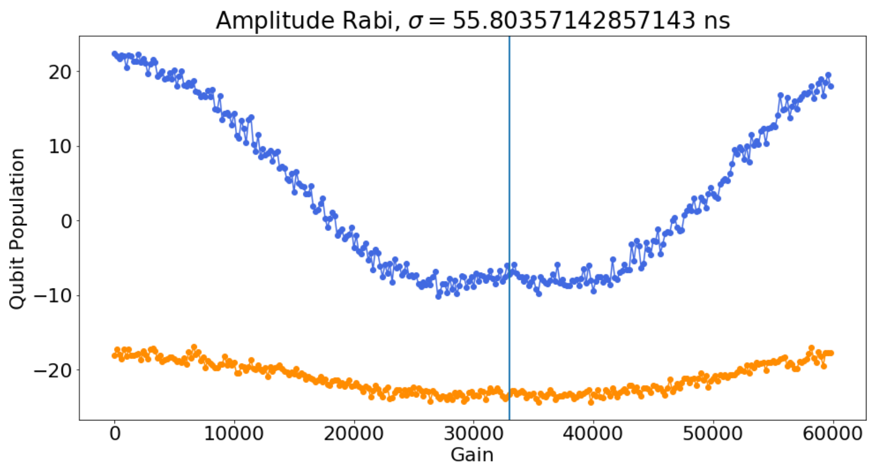

Figure 4: Location of the π pulse in the amplitude Rabi experiment.

4 Single-shot readout: separation of $|g\rangle$ and $|e\rangle$

The `SingleShotProgram` is used to calibrate dispersive readout by experimentally distinguishing the two qubit states.

The idea is to prepare successively:

- the ground state $|g\rangle$;
- the excited state $|e\rangle$ (by applying a π pulse).

For each shot, one obtains a point (I, Q) in the IQ plane. This creates two clouds of points, one for each state.

IQ rotation and threshold

The two IQ clouds are usually tilted in the plane. One therefore applies a rotation of the readout axis:

$$I' = I \cos\theta + Q \sin\theta,$$

so that the two state distributions are as well separated as possible along a single axis.

Once this rotation is applied, the threshold is taken *only on* I'. In other words, the threshold is vertical in the rotated frame, which corresponds to an inclined line in the original IQ plane.

This measurement provides:

- the optimal rotation angle;
- the discrimination threshold;
- the single-shot readout fidelity.

These parameters are then reused in the following experiments to convert the analog signal into a probability of being in $|g\rangle$ or $|e\rangle$.

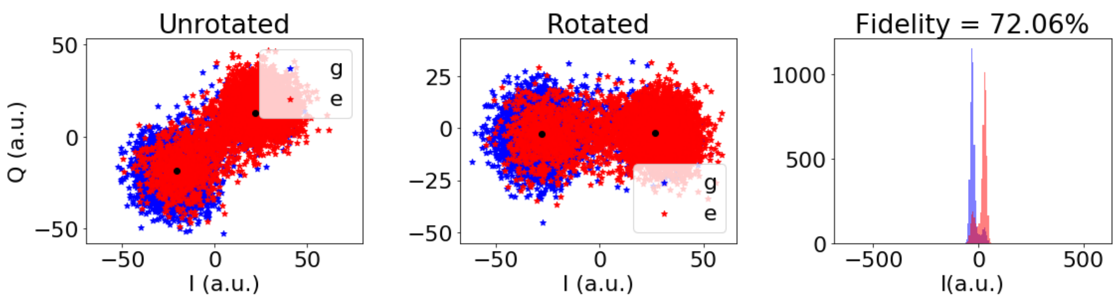

Figure 5: Single-shot measurement of the ground and excited states.

5 Effective qubit temperature

Using the `SingleShotProgram`, one can also estimate the effective qubit temperature using the Boltzmann law.

The qubit is left in the ground state $|g\rangle$, and the shots classified by the threshold as ground-like or excited-like are counted. This gives the probabilities P_g and P_e of being in $|g\rangle$ or $|e\rangle$.

Assuming thermal equilibrium, one has

$$\frac{P_e}{P_g} = \exp\left(-\frac{h f_{ge}}{k_B T_{\text{eff}}}\right).$$

This relation can be inverted to estimate the effective qubit temperature T_{eff}.

6 Relaxation measurement: T_1

The `T1Program` prepares the qubit in the excited state with a π pulse, then lets the system evolve for a variable delay τ before reading it out.

The excited-state population decays according to:

$$P_e(\tau) = Ae^{-\tau/T_1} + B.$$

Physical interpretation

T_1 is the energy-relaxation time of the qubit. It characterizes the probability for the qubit to lose its energy through interactions with the environment.

The fit allows T_1 to be extracted from the decay of the excited-state population.

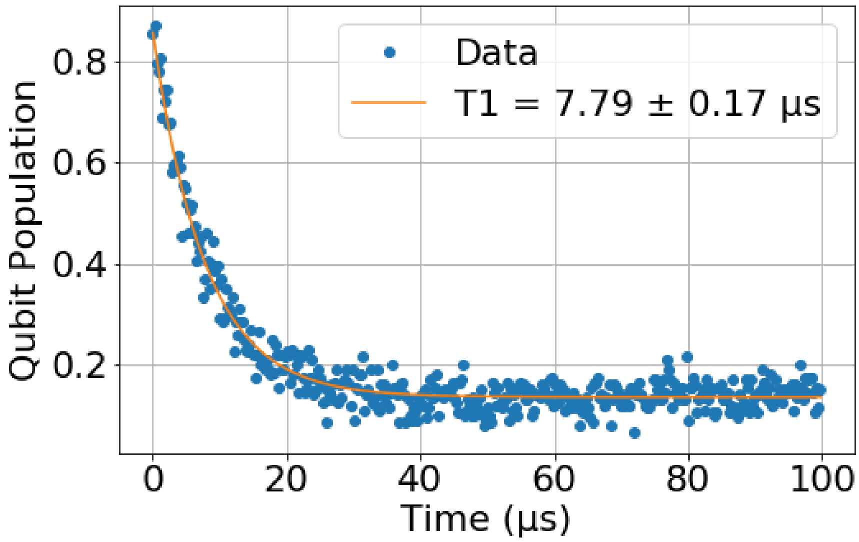

Figure 6: Measurement of T_1.

7 Ramsey coherence measurement: T_2^*

The `RamseyProgram` applies two $\pi/2$ pulses separated by a variable delay τ.

The first pulse prepares a coherent superposition. During the delay, the relative phase between $|g\rangle$ and $|e\rangle$ evolves. The second pulse converts this phase into a measurable population.

The standard fit is:

$$P_e(\tau) = A + B\, e^{-\tau/T_2^*} \cos(2\pi f_R \tau + \phi).$$

Physical interpretation

The Ramsey experiment measures the qubit phase coherence. The oscillations correspond to phase precession, while the exponential envelope reflects the loss of phase coherence due to slow noise, frequency fluctuations, and environmental drifts.

The frequency observed in the Ramsey fringes depends on the detuning between the drive frequency and the qubit resonance, or on the phase advance imposed from one experiment point to the next.

The extracted parameter is:

$$T_2^*.$$

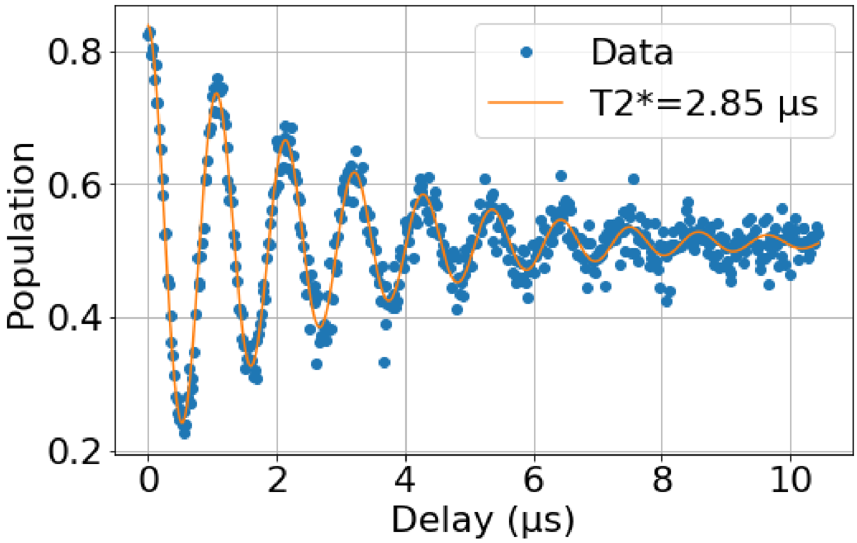

Figure 7: Measurement of T_2^*.

8 Echo coherence measurement: $T_{2,\mathrm{Echo}}$

The `RamseyEchoProgram` follows the Ramsey logic, but inserts a π pulse in the middle of the sequence:

$$\pi/2 \;-\; \tau/2 \;-\; \pi \;-\; \tau/2 \;-\; \pi/2.$$

This central pulse reverses the phase accumulation caused by slow fluctuations.

Purpose

Echo refocuses a large part of the low-frequency noise. One therefore usually observes a longer coherence time than in Ramsey:

$$T_{2,\mathrm{Echo}} \geq T_2^*.$$

The fit can have the same form as in Ramsey if one keeps phase fringes, or a simple exponential envelope if no additional phase ramp is imposed.

Physical interpretation

The echo experiment makes it possible to distinguish slow-noise effects from fast-noise effects. If the envelope decays much more slowly than in Ramsey, it means that the dominant noise is mainly quasi-static or low-frequency.

9 Active reset

Active reset uses the single-shot threshold to decide whether the qubit is in $|g\rangle$ or $|e\rangle$ after an initial measurement.

The experimental protocol is:

1. measure the qubit state;
2. if the readout indicates $|e\rangle$, apply a π pulse;
3. perform a new measurement to verify that the qubit has been brought back to the desired state.

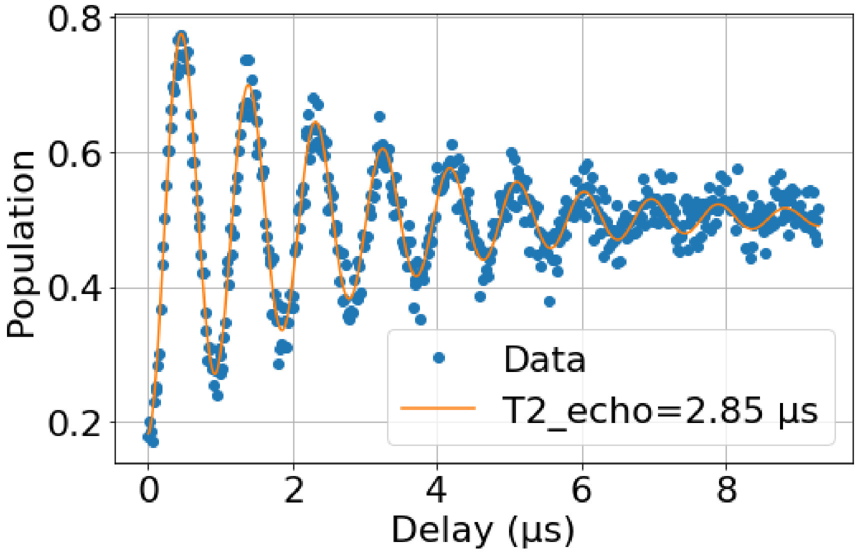

Figure 8: Measurement of $T_{2,\mathrm{Echo}}$.

Usefulness

This step is not primarily about fundamental physics; it is mainly a practical verification that the qubit can be quickly and reliably returned to the desired initial state. This makes repeated experiments much faster, without waiting passively for several T_1 times.

10 QND measurements: double readout and conditional histograms

The last step consists in performing two consecutive readouts without modifying the qubit in between. The first and second measurements are then compared.

One typically plots:

- the distribution of the first readout;
- the distribution of the second readout;
- the second readout conditioned on the first one being $|g\rangle$;
- the second readout conditioned on the first one being $|e\rangle$.

Interpretation

A good QND measurement should preserve the measured state as much as possible:

- if the first readout gives $|g\rangle$, the second one should remain mostly in $|g\rangle$;
- if the first readout gives $|e\rangle$, the second one should remain mostly in $|e\rangle$.

This experiment therefore quantifies both the readout fidelity and its non-demolition character.

Summary

The notebook follows a progressive experimental logic:

1. verify the readout and choose the correct readout parameters;
2. find the qubit frequency;

3. calibrate the π and $\pi/2$ pulses;
4. calibrate the single-shot readout;
5. estimate the effective qubit temperature;
6. measure T_1;
7. measure T_2^* with Ramsey;
8. measure $T_{2,\mathrm{Echo}}$ with Echo;
9. implement active reset;
10. check the QND character of the readout.

Each step prepares the next one: calibrated frequencies are used in the control pulses, single-shot readout is used for thresholding and population measurements, and coherence experiments reveal the physical limits of the qubit.